MAHALO!

Aloha

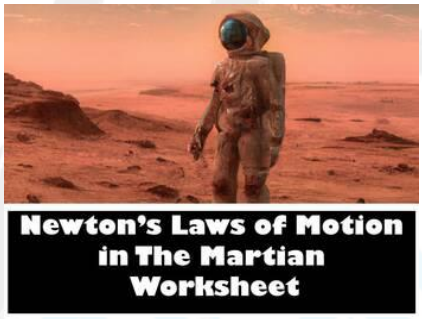
I am thrilled to offer you these worksheets. My email address is educationwithaloha@gmail.com.

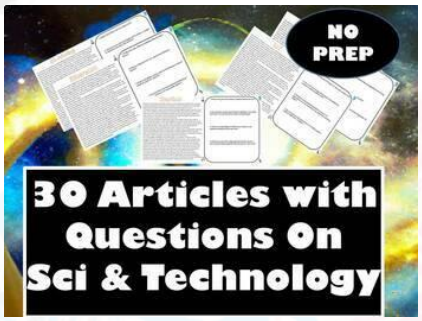
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If you and your students enjoy this, a review would be greatly appreciated. Mahalo for choosing Education with Aloha!

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Instructions

1. Divide students into pairs or small groups.
2. Distribute paper copies of experiment instructions and tables to each group.
3. Guide students to access the simulation on their devices.
4. Instruct students to follow the instructions for Experiment 1
5. After Experiment 1, lead a brief class discussion on the observed relationship.
6. Instruct students to follow the instructions for Experiment 2.
7. After Experiment 2, lead a brief class discussion on the observed relationship.
8. Here is the link to digital versions of worksheets.

I hope you find this resource valuable and engaging for your students. I wish you all the best in your teaching journey and for the rest of the school year.

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Title: Exploring Electrical Circuits: Series and Voltage-Current Relationship

Objectives:

- Students will understand the effect of adding resistors in series on the total resistance and current in a circuit.
- Students will investigate the relationship between voltage and current in a circuit.
- Students will apply Ohm's Law to analyze and explain their experimental findings.
- Students will collect data, create graphs, and analyze the results of their experiments.

Warm-Up:

- Begin the class with a review of the basic components of an electrical circuit (battery, resistor, wire, and switch).
- Introduce the concept of series circuits and discuss the characteristics of a series circuit, such as current and voltage distribution.
- Explain Ohm's Law (V = I * R) and its importance in understanding the behavior of electrical circuits.

Direct Instruction:

- Show a Ted Ed video or a Khan Academy video on electrical circuits to reinforce concepts and provide a visual representation of the material.
- Introduce the two experiments: "Building a Simple Series Circuit" and "Exploring the Relationship between Voltage and Current in a Circuit." Explain the objectives, hypotheses, and variables of each experiment.
- Demonstrate how to set up the circuits and take measurements using the Circuit Construction Kit AC Virtual Lab simulation.
- Explain how to record data in the tables and how to create graphs using graphing paper or Google Sheets.

Assessment:

- Students will complete the two experiments, record their findings in tables, and create graphs to analyze the results.
- Students will answer short-answer questions and challenge questions related to each experiment to demonstrate their understanding of the concepts.
- Collect students' experiment data, graphs, and short-answer questions for evaluation.

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Closure:

- Review the main concepts learned during the experiments, such as the effect of adding resistors in series on total resistance and current, and the relationship between voltage and current in a circuit.
- Discuss the real-world applications of controlling resistance and voltage in circuits.
- Encourage students to think about how these concepts are used in everyday devices and systems.

Tips for Flexibility:

- Adapt the complexity of the experiments depending on students' abilities and prior knowledge of electrical circuits.
- Provide additional support and guidance for students who need assistance with using the simulation, recording data, or creating graphs.
- Encourage students to work in pairs or small groups to facilitate peer learning and support.

Additional Resources:

- TED-Ed: How do batteries work? - Adam Jacobson: https://www.youtube.com/watch?v=9ovNrBW8azM
- Khan Academy: Introduction to circuits: https://www.khanacademy.org/science/electrical-engineering/introduction-to-ee/intro-to-ee/v/circuits-part-1
- IFLScience: The Electric Universe Theory Explained: https://www.iflscience.com/space/the-electric-universe-theory-explained/
- TED-Ed: How do solar panels work? - Richard Komp: https://www.youtube.com/watch?v=xKxrkht7CpY
- Khan Academy: Ohm's Law: https://www.khanacademy.org/science/electrical-engineering/introduction-to-ee/intro-to-ee/v/ee-circuit-analysis-topic-2-1

Standards:

- HS-PS2-9(MA). Analyze simple electrical circuits to predict changes to voltage, current, or resistance when simple changes are made to a circuit.

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Experiment 1: Building a Simple Series Circuit

Objective: To understand how the components in a series circuit interact and affect the flow of current.

Hypothesis: If the number of components (resistors) in a series circuit increases, then the total resistance in the circuit will increase, causing the current to decrease.

Variables:



Independent variable: Number of resistors in the circuit
Dependent variable: Total resistance and current in the circuit
Controlled variables: Voltage source, type and value of resistors

Instructions:

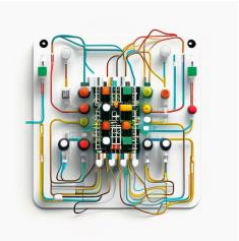
1. Open the Circuit Construction Kit AC Virtual Lab: https://phet.colorado.edu/en/simulations/circuit-construction-kit-ac-virtual-lab
2. Select the "Lab" option.
3. From the right-side panel, drag a battery, a switch, a wire, and a resistor into the workspace.
4. Connect the components to create a simple series circuit: connect the battery to the switch, the switch to the resistor, and the resistor back to the battery using the wires. Check Labels, and Values.
5. Add an ammeter in series with the circuit to measure the current.
6. Record the resistance value of the resistor.
7. Close the switch and note the current reading on the ammeter.
8. For each additional trial, add another resistor in series with the circuit and an additional ammeter to measure the current after each resistor. Be sure to use the same value resistor for each trial.
9. Repeat steps 7-8 four more times, for a total of five trials.

Trial	# of Resistors	Total Resistance (Ω)	Current (A)
1			
2			
3			
4			
5			

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Experiment 1: Building a Simple Series Circuit

Short Answer Questions:



1. How does adding more resistors in a series circuit affect the total resistance?

2. How does adding more resistors in a series circuit affect the current?

3. What happens to the voltage across each resistor as you add more resistors in series?

4. Explain why the changes in total resistance and current occur as you add more resistors in series.

5. How do the results of this experiment support or refute your hypothesis?

Challenge Questions:

1. Calculate the voltage across each resistor in each trial. How does the voltage distribution change with the number of resistors? Use Ohm's Law (V=IR). Do all your calculations check out to the recordings?

2. Create a parallel circuit with the same resistors from the series circuit experiment. Measure the total resistance and current in the parallel circuit. How do these values compare to those from the series circuit? Explain why this occurs.

3. Calculate the power dissipation across the resistors in the series circuit from the experiment. Then, create a new circuit with the resistors in parallel, measure the current through each resistor, and calculate the power dissipation for each resistor. Compare the power dissipation between the two configurations and explain your findings.

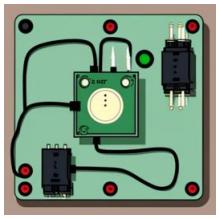
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Experiment 2: Exploring the Relationship between Voltage and Current in a Circuit

Objective: To investigate the relationship between the voltage supplied to a circuit and the current flowing through the circuit.

Hypothesis: If the voltage across a resistor increases, then the current flowing through the circuit will also increase.

Variables:



Independent variable: Voltage supplied to the circuit
Dependent variable: Current flowing through the circuit
Controlled variables: Resistance value, type of resistor

Instructions:

1. Open the Circuit Construction Kit AC Virtual Lab: https://phet.colorado.edu/en/simulations/circuit-construction-kit-ac-virtual-lab
2. Select the "Lab" option.
3. From the right-side panel, drag a battery, a wire, and a resistor into the workspace.
4. Connect the components to create a simple circuit: connect the battery to the resistor, and the resistor back to the battery using the wires.
5. Add an ammeter in series with the circuit to measure the current.
6. Record the resistance value of the resistor.
7. Note the voltage of the battery and the current reading on the ammeter. (Start at 9 Volts)
8. Adjust the voltage of the battery by 1 volt increments, noting the new current reading each time. Repeat this four more times for a total of five trials. Use Ohm's Law (V=IR) to check your recordings.

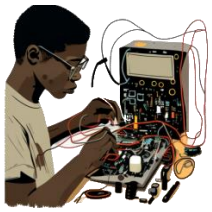
Trial	Voltage (V)	Resistance (Ω)	Current (A)
1	9		
2	10		
3	11		
4	12		
5	13		

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Experiment 2: Exploring the Relationship between Voltage and Current in a Circuit

Short Answer Questions:

1. How does changing the voltage of the battery affect the current flowing through the circuit?



2. What is the relationship between voltage and current in a circuit?

3. According to Ohm's Law (V = I * R), how is the voltage, current, and resistance related in a circuit?

4. Based on your findings, does the relationship between voltage and current support Ohm's Law (V = I * R)? Explain your reasoning.

5. How do the results of this experiment support or refute the hypothesis?

Challenge Questions:

1. Calculate the power dissipated by the resistor in each trial. How does the power change as the voltage increases?

2. If you were to keep the voltage constant and change the resistance, how would the current in the circuit be affected? Explain your reasoning using Ohm's Law (V = I * R).

3. Design an experiment to test the relationship between the resistance in a circuit and the current flowing through the circuit while keeping the voltage constant. What would be your hypothesis, and what variables would you need to control?

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Experiment 1: Building a Simple Series Circuit

ANSWER KEY

Trial	# of Resistors	Total Resistance (Ω)	Current (A)
1	1	10	0.90
2	2	20	0.45
3	3	30	0.30
4	4	40	0.22
5	5	50	0.18

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Experiment 1: Building a Simple Series Circuit
ANSWER KEY

Short Answer Questions:

1. How does adding more resistors in a series circuit affect the total resistance?
Answer: Adding more resistors in a series circuit increases the total resistance. The total resistance in a series circuit is the sum of the individual resistances.

2. How does adding more resistors in a series circuit affect the current?
Answer: Adding more resistors in a series circuit decreases the current. As the total resistance increases, the current flowing through the circuit decreases according to Ohm's Law (V = I * R).

3. What happens to the voltage across each resistor as you add more resistors in series?
Answer: As more resistors are added in series, the voltage across each resistor will vary depending on its resistance. The total voltage supplied by the battery will be divided among the resistors in proportion to their individual resistances.

4. Explain why the changes in total resistance and current occur as you add more resistors in series.
Answer: The changes in total resistance and current occur as more resistors are added in series because the total resistance increases with the addition of each resistor. According to Ohm's Law (V = I * R), as the total resistance increases, the current flowing through the circuit decreases.

5. How do the results of this experiment support or refute your hypothesis?
Answer: The results of this experiment support the hypothesis, as the total resistance increases and the current decreases when more resistors are added in series.

Challenge Questions:

1. Calculate the voltage across each resistor in each trial. How does the voltage distribution change with the number of resistors? Use Ohm's Law (V=IR). Do all your calculations check out to the recordings?
Answer: The voltage across each resistor in a series circuit is proportional to its resistance. As more resistors are added, the voltage distribution changes, with each resistor receiving a portion of the total voltage supplied by the battery.

2. Create a parallel circuit with the same resistors from the series circuit experiment. Measure the total resistance and current in the parallel circuit. How do these values compare to those from the series circuit? Explain why this occurs.
Answer: In a parallel circuit, the total resistance is lower than in a series circuit, and the current is higher. This occurs because the resistances of the individual resistors are combined inversely, resulting in a lower total resistance. According to Ohm's Law (V = I * R), a lower total resistance leads to a higher current.

3. Calculate the power dissipation across the resistors in the series circuit from the experiment. Then, create a new circuit with the resistors in parallel, measure the current through each resistor, and calculate the power dissipation for each resistor. Compare the power dissipation between the two configurations and explain your findings.
Answer: The power dissipation across the resistors in a series circuit will generally be lower than in a parallel circuit. In a series circuit, the current is the same through each resistor, so the power dissipation is calculated using P = I^2 * R for each resistor. In a parallel circuit, the current through each resistor is different and generally higher, leading to higher power dissipation. The power dissipation for each resistor in a parallel circuit is calculated using P = I^2 * R, and the total power dissipation is the sum of the individual power dissipations. The differences in power dissipation are due to the differences in current and resistance between the two configurations.

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Experiment 2: Exploring the Relationship between Voltage and Current in a Circuit

ANSWER KEY

Trial	Voltage (V)	Resistance (Ω)	Current (A)
1	9	10	.90
2	10	10	1
3	11	10	1.10
4	12	10	1.20
5	13	10	1.30

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Experiment 2: Exploring the Relationship between Voltage and Current in a Circuit

ANSWER KEY

Short Answer Questions:

1. How does changing the voltage of the battery affect the current flowing through the circuit?
Answer: As the voltage of the battery increases, the current flowing through the circuit also increases.

2. What is the relationship between voltage and current in a circuit?
Answer: The relationship between voltage and current in a circuit is directly proportional, according to Ohm's Law.

3. According to Ohm's Law (V = I * R), how is the voltage, current, and resistance related in a circuit?
Answer: According to Ohm's Law (V = I * R), the voltage (V) in a circuit is equal to the product of the current (I) and the resistance (R).

4. Based on your findings, does the relationship between voltage and current support Ohm's Law (V = I * R)? Explain your reasoning.
Answer: Yes, the relationship between voltage and current supports Ohm's Law (V = I * R) because the current increases linearly with an increase in voltage, while the resistance remains constant.

5. How do the results of this experiment support or refute the hypothesis?
Answer: The results of this experiment support the hypothesis, as the current flowing through the circuit increases with an increase in voltage.

Challenge Questions:

1. Calculate the power dissipated by the resistor in each trial. How does the power change as the voltage increases?
Answer: The power (P) dissipated by the resistor can be calculated using the formula P = V * I. As the voltage increases, the power dissipated by the resistor also increases.

2. If you were to keep the voltage constant and change the resistance, how would the current in the circuit be affected? Explain your reasoning using Ohm's Law (V = I * R).
Answer: If the voltage remains constant and the resistance increases, the current in the circuit would decrease. This is because, according to Ohm's Law (V = I * R), the current (I) is inversely proportional to the resistance (R) when the voltage (V) is constant.

3. Design an experiment to test the relationship between the resistance in a circuit and the current flowing through the circuit while keeping the voltage constant. What would be your hypothesis, and what variables would you need to control?
Answer:
Hypothesis: If the resistance in a circuit increases while keeping the voltage constant, the current flowing through the circuit will decrease.

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